Aircraft Tutorial

"The effects of radical propulsive solutions on the characteristics of regional aircraft"

1. Introduction

1.1.Instructions

This document provides all necessary information you will need to prepare and to execute the Aircraft Tutorial. Read this entire document carefully before beginning to work on it.
It is assumed that you have studied the lecture notes of the course with a special attention to the aircraft design specific module dedicated to **weight&balance** and **stability&control**. Some of the additional material present on the book suggested in as a study material and related to the aforementioned topics, can be useful to perform this tutorial, hence you are advised to familiarize with it. It is assumed that you are able to use knowledge from the previous course AE2111 (wing design modules).

1.2.Project background and objective

The Bombardier CRJ1000 is a regional jet airliner that was designed and manufactured by Canadian transportation conglomerate Bombardier (formerly Canadair) between 1999 and 2020. The CRJ program was acquired by Japanese corporation Mitsubishi Heavy Industries in 2020, which ended production of the aircraft due to the market saturation with similar competitors and more efficient turboprop alternatives.
From both operators (airliners, pilots) and passengers, this aircraft is still very appreciated for its flight characteristics and comfort, as well as for its low purchasing costs.

An IBERIA CRJ-1000 [copyright @ K. von Wedelstaedt]

AeroSpark is a recent European engineering company specialized in the modification of out-of-production aircraft which have the potential to be transformed into new competitive aircraft, by means of new propulsive solutions and systems and ensuring at the same time limited re-design processes. *AeroSpark* has just started a joint programme with Mitsubishi to evaluate the potential benefits of re-designing the CRJ-1000 and to eventually propose a new **CRJ-EXX** version equipped with a radically innovative Hybrid-Electric propulsion system (whose some specifications, that impact your study, are reported in the following paragraphs).

Mission
You and your 5 colleagues, are working in the *Stability&Control* department of *AeroSpark* and you support the other departments with the aim to finalize the first conceptual assessment of the new CRJ-EXX configuration. Your team has been asked to perform a design investigation and report it.
In particular your group has to deliver a report composed by two studies: the general instructions for these activities are provided in the next paragraphs of the present document.

2. First Part: assessment of the reference aircraft

2.1. Deliverables of the first part

The characteristics of the reference aircraft need to be determined prior to evaluate possible further modifications. Therefore, your group is asked to determine weights, balance, stability and controllability parameters of the reference CRJ-1000 configuration with **with full economy, 100 pax @ 31 inch seats, 2+2 abreast**.
In particular you are asked to:

a) **Familiarize** with the entire task list and **search for relevant data and information** on the CRJ-1000 aircraft (e.g., general dimensions; weights and performance data; layout of cabin, wing, tail, high lift devices, landing gear, etc.). For the retrieval of data, you can consult website providing original manufacturer specifications and/or the Jane's All the World's Aircraft catalogue (available on paper at the TUD library or digitally at www.Janes.com, when accessed within the TU network). Other popular websites provide plenty of data especially in the form of "Airport Planning Manual" document.

b) **Generate a table and a pie chart** containing the values of Maximum Takeoff Weight (MTOW), Operational Empty Weight (OEW), maximum payload weight and fuel weight@max payload. The payload weight will be further split into 1) pax&cabin luggage weight, 2) front cargo hold weight and 3) aft cargo hold weight. The table shall indicate the values of OEW, fuel weight and maximum payload weight both in absolute values and as percentage of MTOW.

c) The weight department has provided the following Table 1, containing the breakdown (expressed **in percentage of the EOW**) of the main aircraft systems:

Table 1: weight breakdown

Component	Weight (% EOW)
Wing	19.7
Horizontal Tail	2.5
Vertical Tail	1.8
Fuselage (including cabin and furnishing systems)	35.0
Main Landing gear	5.8
Nose Landing gear	0.8
Propulsion System (including Nacelles)	13.3
Cockpit Systems (avionics, furnishing…)	2.3

Note some of the items composing the Empty Operative configuration might be excluded from the table. In that case, you can assume that the components that are not reported in the table simply don't affect notably the position of the CG. **Generate a table** where information of the different centers of gravity are reported according the following guidelines:

I. The components shall be re-arranged into a "fuselage group" and a "wing group" (make sure to include the right components in each group!). The total weigh of both groups shall be provided as well.

II. Calculate the CG of groups and components; you shall make use of **two reference systems**:
   - One global system positioned ahead of the aircraft;
   - One system fixed at LEMAC (i.e., the leading edge of the wing mean aerodynamic chord).

III. In the table you will report the CG positions of the components, of both the wing and fuselage groups and of the empty operative configuration of the aircraft with respect to reference system fixed at LEMAC. Express these CG positions as percentage of the wing mean aerodynamic chord (MAC)

d) Consequently **generate the aircraft loading diagram**. Organize the passengers into groups, according to their seating position following the window-aisle rule. Place the remaining payload weight in the front and aft cargo holds, according to their capacity. Make use of drawings and payload arrangement as available on the websites of manufacturer and/or operators.

e) **Identify the most aft and forward c.g. positions** during operation and report them in a table, indicated as percentage of the aircraft mean aerodynamic chord MAC. Indicate in the same table the specific loading conditions that are responsible for the achievement of the most aft and forward c.g. positions. In the same table, you will report also the position of the aircraft c.g.@OEW (indicated as percentage of MAC), the c.g. position of both the front and the rear cargo, and the c.g. position of the fuel tank(s).

f) Determine and **report** in a table **all the geometry and aerodynamics** parameters (see point e) in the next Part2) that are necessary to assess both stability and controllability characteristics.

g) Ultimately **generate the complete scissor plot** of the reference aircraft with the neutral stability line, the stability line and the controllability line.

3. Second Part: evaluation of design modifications

In the second part of your study, you are asked to evaluate possible effects of some design input coming directly from the other company departments.

3.1.Customer design modifications

AeroSpark has already performed some design work towards the development of the CRJ-EXX. The CRJ-EXX will feature new lightweight interiors, a new glass cockpit and, in particular, **a new <u>serial hybrid electric propulsion system;</u>** its arrangement consists of two modern high bypass ratio turbofans coupled with electric motors that provide boost power during demanding flight phases. Those systems are arranged together and placed in a nacelle located at the same place of the older ones. In addition, the other departments of *AeroSpark* have estimated the following other modifications:

I. The nacelles have increased dimension both in length (+20%) and diameter (+20%).
II. Despite the increase in size, the propulsion System doesn't present variation in neither weight or location of its c.g..
III. Due to the utilization of innovative material, the wing weight is reduced by 9% with respect to its initial absolute value, and the fuselage weight is reduced by 7% (again with respect to its initial absolute value).
IV. The electric part of the powertrain is powered by two battery packs. The battery total weight is 4000 kg and the first pack (that accounts for a total volume of 1300 liters and a weights of 2025 kg) is located in the most forward part of the forward cargo hold. The second battery pack (2475 kg, and a volume of 1600 liters) is instead located in the aft cargo compartment, in the most aft area.
V. The weights (and hence the CG) of the batteries are not considered in the EOW configuration reported in point I. Therefore, a new CG@EOW+Batt shall be calculated since <u>the batteries are not removable.</u>
VI. The last three rows of passenger are removed.
VII. The cargo weight is remaining equal to the value of the old CRJ-100 but its distribution is proportional to the new volumes of the cargo holds (reduced because of the presence of the batteries).
VIII. The **MTOW** of the CRJ-EXX will be **the same** of the CRJ-1000; therefore, the modifications of the weights terms (payloads and EOW+Batteries) are compensated by a proper modification of the amount of fuel embarked.
IX. The Aerodynamic is improved with **new winglet system** that increases the wing effective Aspect Ratio by 25% when compared to the initial CRJ-1000.
X. The maximum lift coefficient of the Aircraft-less-tail is increased by 20% with respect the CRJ-1000 version, thanks to improvements of the High-Lift Devices.
XI. In order to guarantee safe and more effective recharging operations, the height of the landing gears is increased by 30 cm.

3.2. Customer questions

Phoenix Works is asking your team to address the following question:

a) *How do the aforementioned design modifications affect both longitudinal stability and controllability? Provide evidence of your answer providing calculation steps where necessary.*
b) *Would it be still possible to fulfill the longitudinal S&C requirements without changing the current horizontal tail size, longitudinal wing position and stability margin?*

If **NOT:**

c) *What would be the critical longitudinal requirement? Stability or controllability?*
d) *How larger should be the horizontal tail to fulfil again both the longitudinal S&C constraints, while maintaining the same longitudinal position of the wing?*
e) *What design iterations (if any) would be necessary to confirm your answer?*

If **YES:**

c) *How smaller could be the horizontal tail size, while still fulfilling both the longitudinal S&C constraints and without modifying the longitudinal position of the wing neither the stability margin?*
d) *In this case what would become the critical longitudinal requirement? Stability or control?*
e) *What design iterations (if any) would be necessary to confirm your answer?*

f) *How do the aforementioned design modifications affect the design of the landing gears according to ground operations? Provide evidence of any changes in the ground requirements with respect to the original CRJ-1000.*

Recommendations:

- Make use of loading diagram and scissor plot to provide clear evidence to your answers.
- If you are suggesting the need of iterations, be specific on the parameters to be iterated.
- Consider that the horizontal tailplane surface of T-tail aircraft, in order to account for deep stall risk, is generally 15% larger than that required to just satisfy the S&C requirements.

3.3. Deliverables of the second part

a) Generate a second **weight table** for the CRJ-EXX aircraft similar to the one indicated in **deliverable b)** of the part1, in this case report also the weight of the batteries as a standalone weight term.
b) Provide an updated **c.g. data table** breakdown table similar to the one aske at point c) of the part 1. Also include the contribution of the two different battery packs.
c) Generate a second **loading diagrams** for the CRJ-EXX aircraft similar to Task d) of part 1. Clearly report both the CG@EOW and the CG@EOW+Batt. Preferably you will display the two loading diagrams together on the same plot. In case they will be displayed in two separate plots, their scale and size must be adequate to compare easily the difference of the two diagrams.

d) Generate **the complete scissor plot** (stability + neutral stability + controllability curves) for the CRJ-EXX reporting also the CG range. The scissor plot must be scaled and sized in such a way it can be easily compared to the one of the reference aircraft. To make this comparison easy, you can report the two scissor plots together in a separate figure.

e) **Provide a table** displaying the data necessary to build the scissor plot(s). You shall indicate at least the following data for both the RJ85 and the RJXX:
   - Cruise speed
   - Approach speed
   - Tail-wing speed ratio (Vh/V)
   - Lift rate coefficient of the tail at cruise speed
   - Lift rate coefficient of the aircraft-less-tail at cruise speed, including the separate contributions of:
     - Wing
     - Fuselage
   - Wing downwash gradient
   - Position of the aerodynamic center of the aircraft-less-tail, evaluated both at cruise and approach speed, and including the following separate contributions:
     - Wing
     - Fuselage
     - Nacelles
   - Maximum horizontal tail lift coefficient (indicate also the type of tail, e.g. full moving, fixed, etc)
   - Zero Lift pitching moment coefficient of the aircraft-less-tail, including the separate contributions of all the aircraft parts.

f) Based on calculations and on the qualitative reasoning, provide an answer to each of the customer questions reported in paragraph 3.2 from *a)* to *f)*

4. Part3: organization of the group

The tutorial is a group work where all the group members must contribute equally. Each student must actively perform both the calculation/design tasks and reporting ones. It is up to the group members to organize the group and to constantly self-monitor progresses and to do possible actions to improve the outcome of the tutorial.
Group organization as well as tasks division shall be briefly reported in an appendix (max 3 pages long) of the report.

5. Requirements on the report and grading criteria

The report must adhere to the following guidelines:

- Your group must deliver an electronic version of the report in PDF format not later than the deadline indicated on BrightSpace for the delivery of the tutorial in the regular session.
- The report will be uploaded in the "Aircraft Tutorial - Regular Session" assignment folder that will be created on BrightSpace by the teaching staff.

- The report shall be named: *"AE3211_Aircraft_Tutorial_Group_#"* (where # indicates your group number).
- The report shall clearly indicate on the front page, the name and the student number of all the team members, as well as the group number assigned during the tutorial session.
- The report has a page limit of 25 pages, all included: it must include all the deliverables described in both Part1, Part 2 as well as the organizational appendix (Part 3).
- In addition to the asked deliverables, a **limited** amount of text will be necessary to answer to the customer questions, describe the main characteristics, justify and state all the design assumptions and decisions. Text to explain theoretical background is not needed.
- Shows all the numerical results with proper unit of measures and shows calculation steps when they are needed to answer the deliverables.
- Include clear and readable plots: do not forget labels and consider to scale its dimension to favor comparison when requested (e.g. comparison of loading diagrams).
- Include a list of references when data and/or equation are used from literature.
- **Ignoring the aforementioned indications, might affect the final grade for this assignment.**

The report will be graded as a whole (no intermediate points for each part will be given) according to the following criteria:

- **Completeness and correctness:** Did you answer all the posed questions? Did you include all the requested deliverables and information? Did you properly compute all the coefficients, at the required conditions? Did you indicate the correct units?
- **Consistency:** Did you use data and assumptions consistently through your report? Do your plots match your assumptions and calculations? Do you explain/justify the source of all your data, plots, charts etc.?
- **Reporting quality**: Is your story clear and concise? Is the report properly structured? Are your paged numbered, as well as your figures, tables and equations? Are all your tables and plots readable and provided with caption, units, labels and legend?